Factors, Multiples and Primes

Name _____ **Date** _____ No calculator Version 1

KEY WORDS factor · multiple · prime number · composite number · highest common factor (HCF) · least common multiple (LCM)

A Factors ●○○ — ANSWER ON THE SHEET

Remember: A factor of a number divides into it exactly. Factors come in pairs: $12 = 1 \times 12 = 2 \times 6 = 3 \times 4$.

1. Write all the factor pairs of each number.

a) 12 _____ **b)** 18 _____

c) 20 _____ **d)** 30 _____

2. List all the factors of each number.

a) 8 _____ **b)** 15 _____

c) 24 _____ **d)** 36 _____

e) 13 _____ **f)** 40 _____

B Multiples ●○○ — ANSWER ON THE SHEET

Watch out: Factors are the numbers that divide into a number. Multiples are the numbers in its times table. The factors of 6 are 1, 2, 3, 6. The multiples of 6 are 6, 12, 18, ...

3. Write the first five multiples of each number.

a) 3 _____ **b)** 7 _____

c) 9 _____ **d)** 12 _____

4. Circle the multiples in each list.

a) Multiples of 4: 6 12 18 20 30 36 42

b) Multiples of 6: 12 16 18 26 30 42 56

c) Multiples of 9: 18 24 27 39 45 54 63

C Prime and composite numbers ●●○ — ANSWER ON THE SHEET

Remember: A prime number has exactly two factors: 1 and itself. A composite number has more than two factors. The number 1 is neither.

5. Circle the prime numbers in each list.

a) 2 9 11 15 21 23 27 29

b) 1 31 33 37 39 41 49 51

c) 53 57 61 63 67 77 83 91

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6. Is each number prime or composite?

a) 17 _____ **b)** 21 _____ **c)** 39 _____ **d)** 43 _____

e) 49 _____ **f)** 2 _____ **g)** 87 _____ **h)** 1 _____

D Highest common factor ●●○

WORK IN YOUR BOOK

> **Remember:** List the factors of both numbers. The HCF is the largest number that is in both lists.

7. Find the highest common factor of each pair.

a) 6 and 9 **b)** 8 and 12 **c)** 10 and 15 **d)** 12 and 18

e) 14 and 21 **f)** 20 and 30 **g)** 16 and 24 **h)** 9 and 20

E Least common multiple ●●○

WORK IN YOUR BOOK

> **Remember:** List the multiples of both numbers. The LCM is the smallest number that is in both lists.

8. Find the least common multiple of each pair.

a) 2 and 3 **b)** 4 and 6 **c)** 3 and 5 **d)** 6 and 9

e) 2 and 7 **f)** 5 and 6 **g)** 4 and 8 **h)** 6 and 8

F Problems ●●●

WORK IN YOUR BOOK

9. One light flashes every 4 seconds and another flashes every 6 seconds. They flash together now. After how many seconds do they next flash together?

10. A teacher has 18 pencils and 24 rubbers. She makes identical packs that use all of them, with no items left over. What is the largest number of packs she can make?

★ Extension

FOR EARLY FINISHERS

11. List all the prime numbers between 20 and 30.

12. Write each number as a product of prime numbers.

a) 6 **b)** 15 **c)** 12

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Factors, Multiples and Primes

Name _____ **Date** _____ No calculator **Version 2**

KEY WORDS factor · multiple · prime number · composite number · highest common factor (HCF) · least common multiple (LCM)

A Factors ●○○ — ANSWER ON THE SHEET

Remember: A factor of a number divides into it exactly. Factors come in pairs: $12 = 1 \times 12 = 2 \times 6 = 3 \times 4$.

1. Write all the factor pairs of each number.

a) 10 _____ **b)** 16 _____

c) 28 _____ **d)** 24 _____

2. List all the factors of each number.

a) 6 _____ **b)** 21 _____

c) 32 _____ **d)** 48 _____

e) 11 _____ **f)** 45 _____

B Multiples ●○○ — ANSWER ON THE SHEET

Watch out: Factors are the numbers that divide into a number. Multiples are the numbers in its times table. The factors of 6 are 1, 2, 3, 6. The multiples of 6 are 6, 12, 18, ...

3. Write the first five multiples of each number.

a) 4 _____ **b)** 6 _____

c) 8 _____ **d)** 11 _____

4. Circle the multiples in each list.

a) Multiples of 3: 9 14 18 22 27 31 36

b) Multiples of 7: 14 21 27 35 44 49 54

c) Multiples of 8: 16 20 24 36 40 56 62

C Prime and composite numbers ●●○ — ANSWER ON THE SHEET

Remember: A prime number has exactly two factors: 1 and itself. A composite number has more than two factors. The number 1 is neither.

5. Circle the prime numbers in each list.

a) 3 8 13 15 19 21 25 31

b) 1 27 37 43 45 47 57 59

c) 51 61 69 71 73 81 87 97

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6. Is each number prime or composite?

a) 19 _____ **b)** 27 _____ **c)** 33 _____ **d)** 41 _____

e) 55 _____ **f)** 2 _____ **g)** 91 _____ **h)** 1 _____

D Highest common factor ●●○

WORK IN YOUR BOOK

> **Remember:** List the factors of both numbers. The HCF is the largest number that is in both lists.

7. Find the highest common factor of each pair.

a) 4 and 6 **b)** 9 and 12 **c)** 10 and 25 **d)** 12 and 16

e) 15 and 21 **f)** 20 and 50 **g)** 18 and 24 **h)** 8 and 15

E Least common multiple ●●○

WORK IN YOUR BOOK

> **Remember:** List the multiples of both numbers. The LCM is the smallest number that is in both lists.

8. Find the least common multiple of each pair.

a) 2 and 5 **b)** 4 and 6 **c)** 3 and 4 **d)** 6 and 8

e) 3 and 7 **f)** 5 and 7 **g)** 3 and 9 **h)** 8 and 9

F Problems ●●●

WORK IN YOUR BOOK

9. One light flashes every 6 seconds and another flashes every 9 seconds. They flash together now. After how many seconds do they next flash together?

10. A teacher has 20 pencils and 30 rubbers. She makes identical packs that use all of them, with no items left over. What is the largest number of packs she can make?

★ Extension

FOR EARLY FINISHERS

11. List all the prime numbers between 30 and 40.

12. Write each number as a product of prime numbers.

a) 10 **b)** 21 **c)** 18

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